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SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

MINI PROJECT REPORT



IPv4 Subnetting Using Cisco Packet Tracer

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ABSTRACT

This mini project focuses on understanding and implementing IPv4 subnetting using Cisco Packet Tracer. The main objective of the project was to divide a single IPv4 /24 network into smaller subnets, configure networking devices, and verify communication between hosts.

A base network of 192.168.10.0/24 was subnetted into two /25 subnets. Networking devices such as a router, switches, and personal computers were configured with appropriate IP addresses, subnet masks, and default gateways. Connectivity between devices was tested using ping commands, and packet flow was analysed using Simulation Mode in Cisco Packet Tracer.

This project provides hands-on experience with subnetting calculations, IP addressing, and routing concepts. The successful completion of the project confirms a clear understanding of basic IPv4 subnetting and practical network configuration.

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1. INTRODUCTION

IPv4 subnetting is a fundamental networking technique used to divide a large network into smaller, manageable networks called subnets. Subnetting helps reduce broadcast traffic, improves network performance, and enhances security.

In modern networks, efficient IP address management is essential. This mini project demonstrates how subnetting can be practically implemented using Cisco Packet Tracer. By dividing a single /24 network into two subnets and configuring networking devices, this project bridges the gap between theoretical knowledge and real-world networking practice.

1.1 OBJECTIVES

The objectives of this mini project are:

- To understand the concept of IPv4 subnetting
- To divide a /24 network into two subnets
- To calculate subnet masks, network addresses, and usable IP ranges
- To configure routers, switches, and PCs
- To verify connectivity between devices using ping
- To observe packet flow using Simulation Mode

2. IPv4 SUBNETTING

Subnetting is the process of borrowing bits from the host portion of an IP address to create additional network addresses. This allows a single network to be split into multiple smaller networks.

Advantages of Subnetting

- Better network management
- Reduced broadcast traffic
- Improved performance
- Enhanced security
- Efficient utilization of IP addresses

In this project, subnetting was used to divide one Class C network into two smaller subnets to enable structured communication.

2.1 SUBNETTING CALCULATIONS

Base Network

- Network Address: **192.168.10.0/24**
- Subnet Mask: **255.255.255.0**
- Network bits: 24
- Host bits: 8

Number of Subnets Required

- Required subnets: **2**

To obtain 2 subnets:

21=22^1 = 221=2

So, **1 bit** is borrowed from the host portion.

New Subnet Information

- New Prefix: **/25**
- New Subnet Mask: **255.255.255.128**
- Block Size:

256−128=128256 - 128 = 128256−128=128

This means each subnet increases by 128 in the last octet.

2.2 SUBNET DETAILS

Subnet A

1. Network Address: **192.168.10.0/25**
2. Usable IP Range: **192.168.10.1 – 192.168.10.126**
3. Broadcast Address: **192.168.10.127**

Subnet B

- Network Address: **192.168.10.128/25**
- Usable IP Range: **192.168.10.129 – 192.168.10.254**
- Broadcast Address: **192.168.10.255**

Each subnet supports **126 usable hosts**.

3. NETWORK DESIGN AND TOPOLOGY

The network consists of:

- 1 Router
- 2 Switches
- 4 PCs

Each subnet is connected to the router through a switch. The router enables communication between the two subnets. All devices were connected using copper straight-through cables**.**

This figure shows the overall network topology consisting of one router, two switches, and four PCs. The topology illustrates how two /25 subnets are connected through the router to enable inter-subnet communication.

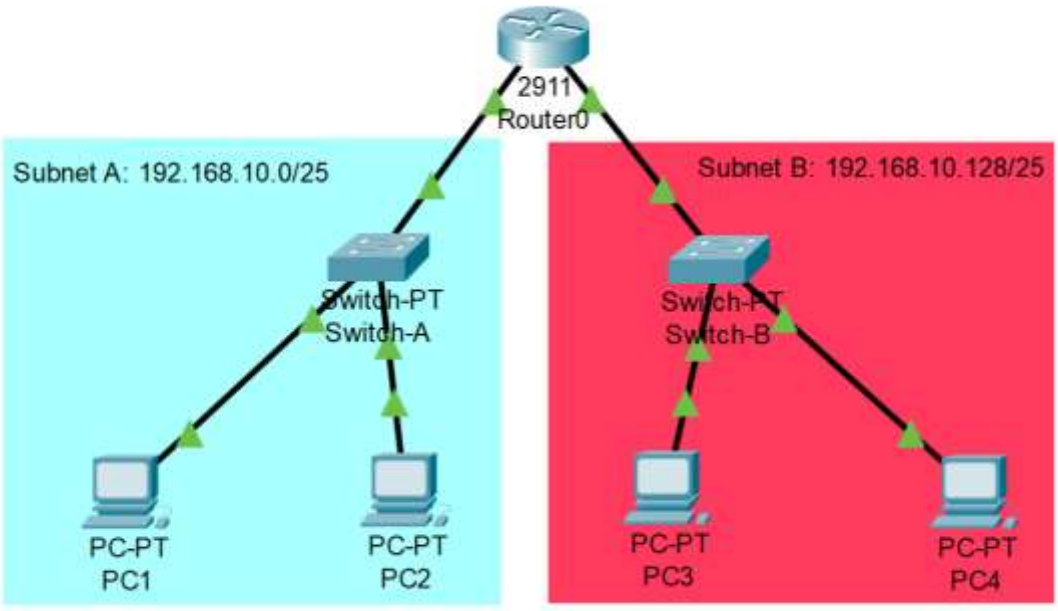


Figure 1:Network Topology of IPv4 Subnetting in Cisco Packet Tracer

4. IP ADDRESSING SCHEME

Device	IP Address	Subnet Mask	Default Gateway
Router G0/0	192.168.10.1	255.255.255.128	—
Router G0/1	192.168.10.129	255.255.255.128	—
PC1	192.168.10.10	255.255.255.128	192.168.10.1
PC2	192.168.10.20	255.255.255.12	192.168.10.1
PC3	192.168.10.130	255.255.255.128	192.168.10.129
PC4	192.168.10.140	255.255.255.128	192.168.10.129

4.1 Router Configuration

- **G0/0:** 192.168.10.1 /25
- **G0/1:** 192.168.10.129 /25

Router Functions:

- Acts as a gateway for both subnets

- Routes traffic between Subnet A and Subnet B

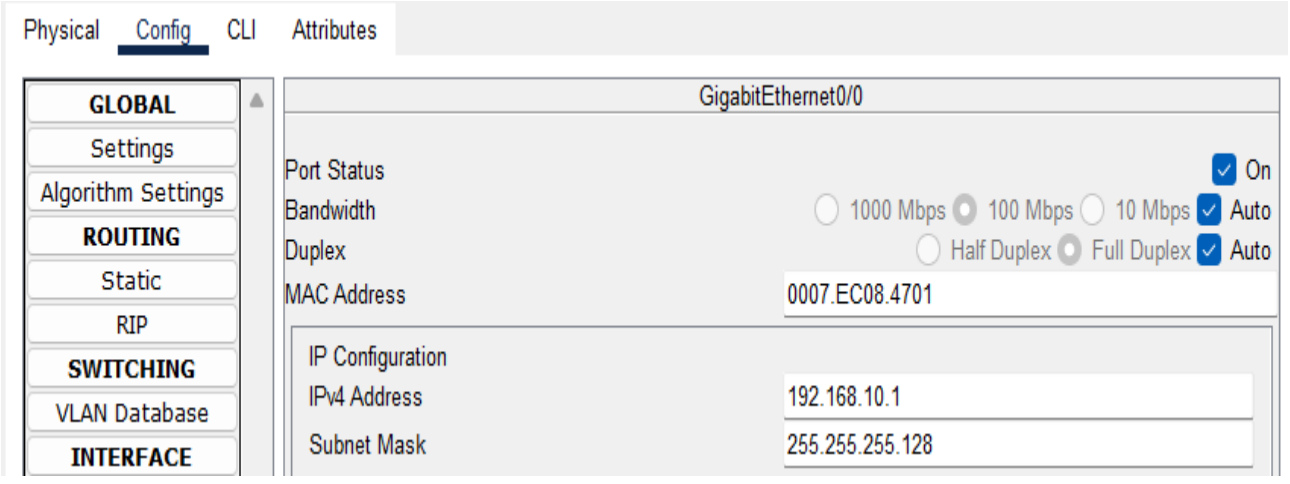


Figure 2: Router G0/0 Configuration

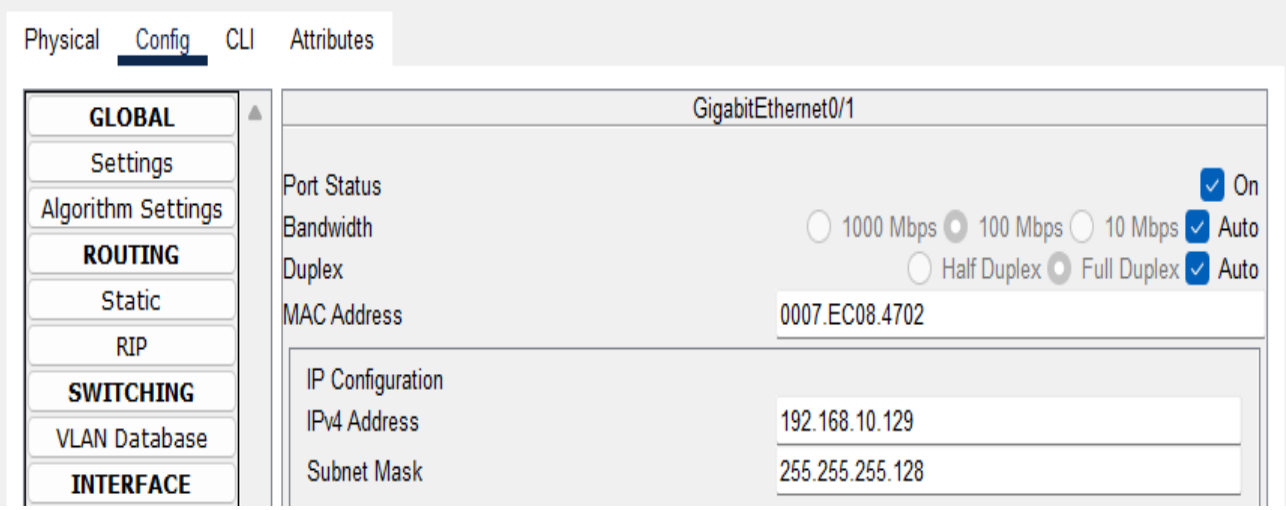


Figure 3: Router G0/1 Configuration

4.2 PC Configuration

Each PC was configured using the IP Configuration option.

Correct IP address, subnet mask, and default gateway were entered.

Subnet A

- PC1: 192.168.10.10 /25 | Gateway: 192.168.10.1
- PC2: 192.168.10.20 /25 | Gateway: 192.168.10.1

Subnet B

- PC3: 192.168.10.130 /25 | Gateway: 192.168.10.129
- PC4: 192.168.10.140 /25 | Gateway:192.168.10.129



Figure 4: PC1 IP Configuration



Figure 5: PC2 IP Configuration



Figure 6: PC3 IP Configuration

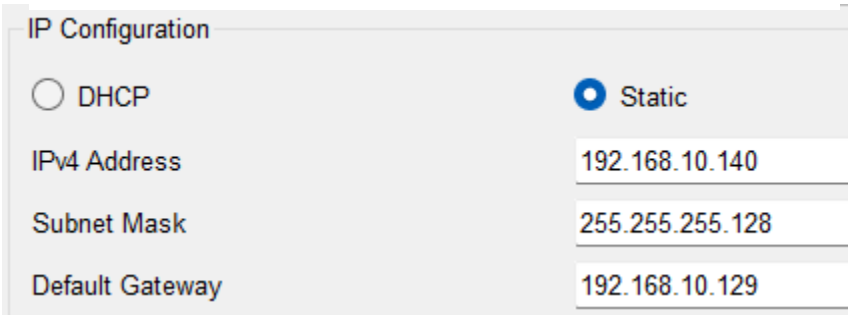


Figure 7: PC4 IP Configuration

The default gateway allows PCs to communicate with devices in other subnets.

5. CONNECTIVITY TESTING AND SIMULATION

5.1 Connectivity Testing (Ping)

Ping tests were performed to verify connectivity:

- PC1 → PC2 (same subnet)
- PC1 → PC3 (different subnet)
- PC1 → PC4 (different subnet)

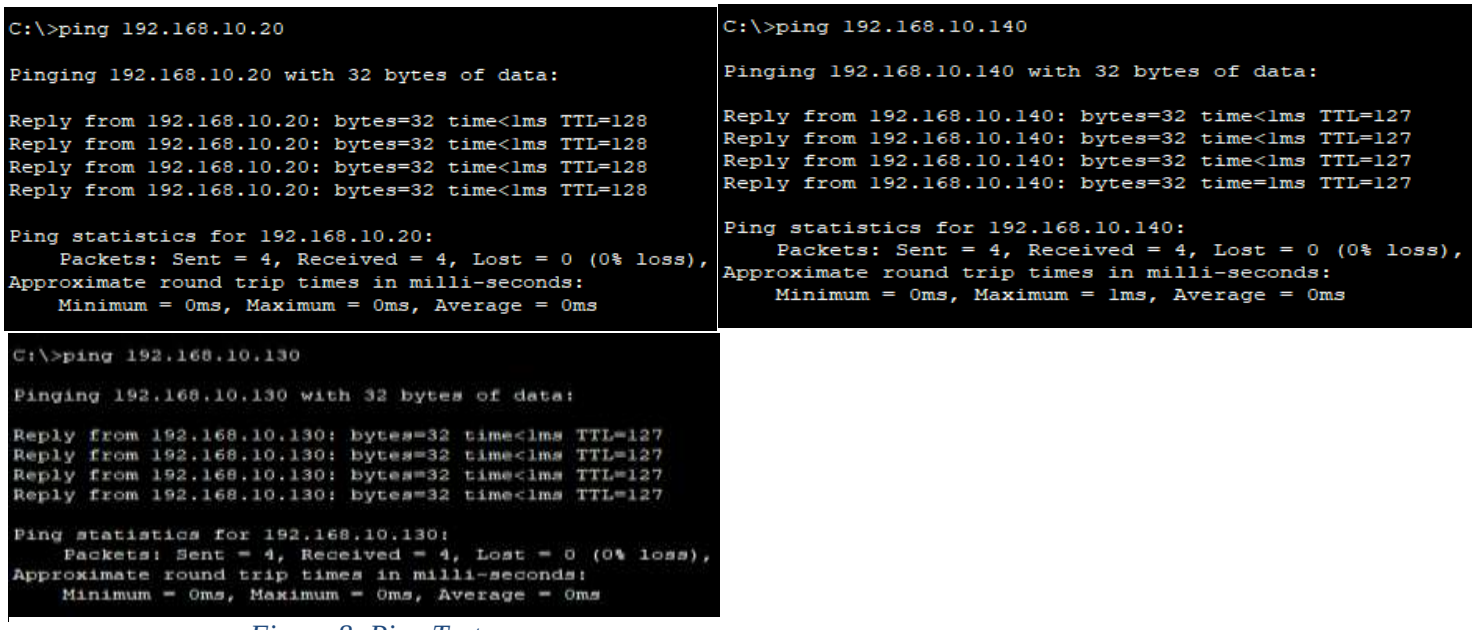


Figure 8: Ping Test

All ping tests were successful, confirming proper network configuration.

This figure shows successful ping replies verifying communication within the same subnet and across different subnets through the router.

5.2 Simulation Mode Analysis

Cisco Packet Tracer Simulation Mode was used to observe packet flow. A Simple PDU was sent from PC1 to PC2 and PC1 to PC3.

Observed packets included:

- ARP Request
- ARP Reply
- ICMP Echo Request
- ICMP Echo Reply

This confirmed successful routing and packet delivery.

This figure illustrates packet transmission between hosts using Simulation Mode, including ARP and ICMP packets.

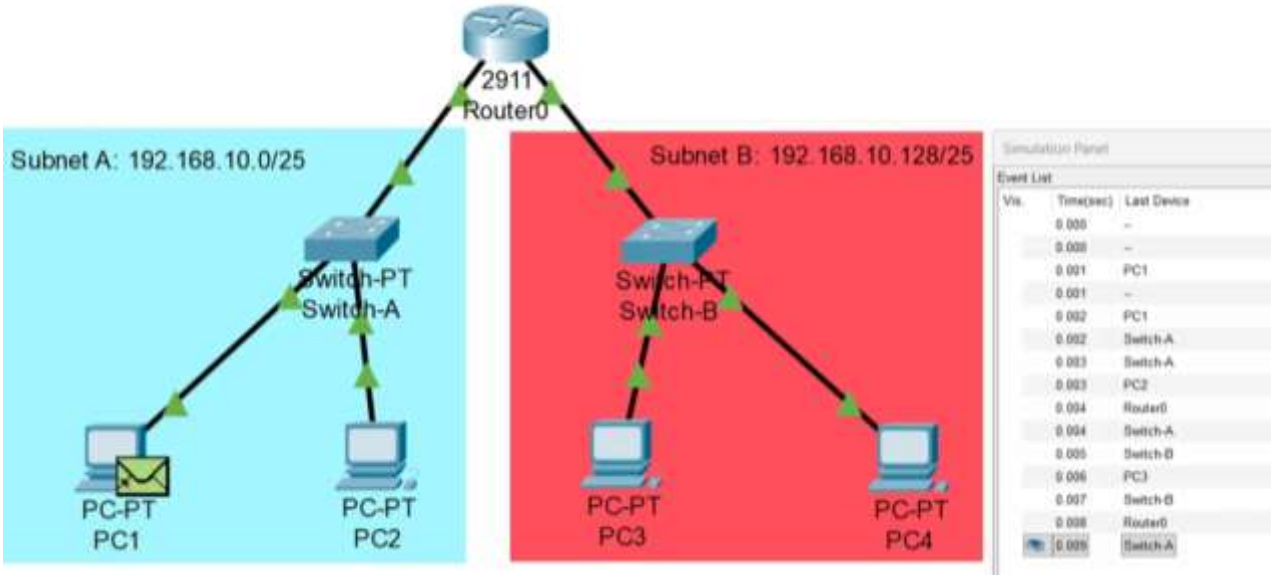


Figure 9: Packet Flow Observation Using Simulation Mode

6. RESULTS AND DISCUSSION

The results of the project showed that:

- Subnetting was performed correctly
- IP addressing was accurate
- Router successfully routed packets
- No packet loss occurred after configuration

Initial ping delay was observed due to ARP resolution, which is normal network behavior.

7. CONCLUSION

This mini project successfully demonstrated the implementation of IPv4 subnetting using Cisco Packet Tracer. A /24 network was divided into two /25 subnets, and all networking devices were configured correctly.

Connectivity was verified using ping tests and Simulation Mode. The project provided practical understanding of subnetting, IP addressing, and routing concepts, forming a strong foundation for advanced networking studies.

REFERENCES

1. Cisco Packet Tracer – IPv4 Subnetting Mini Project Notes
2. Basic IPv4 Subnetting – Step-by-Step Explanation
3. Computer Networking Fundamentals – IPv4 Addressing